

Suryansh Aryan

+1 540-557-8599 | suryaryan@vt.edu | [LinkedIn](#) | [GitHub](#) | [Personal Website](#)

EDUCATION

Kevin T. Krofton Department of Aerospace Engineering, Virginia Tech	Blacksburg, VA, USA
<i>M.S Thesis in Aerospace Engineering, Specialization in Dynamics & Control, CGPA: 3.85/4.00</i>	Aug 2023 – May 2026
Manipal University	Manipal, KN, India
<i>B.Tech in Aerospace Engineering, Minor in Computational Mathematics CGPA: 9.09/10.00</i>	July 2019 – July 2023

EXPERIENCES

Lead Graduate Research Assistant	May 2024 – Present
<i>Center for Space Science and Engineering, Virginia Tech</i>	<i>Blacksburg, VA, USA</i>
- Spearheaded NSF-funded project development as part of a space engineering team, driving LEO satellite constellation ground software solutions for near-Earth and deep space communication resiliency. - Developed and maintained Virginia Tech's in-house deep space emulator SpaceNet, a MiniNet-based network topology and mission operations testbed, outperforming state-of-the-art emulators like Hypatia and NS3. - Led development of realistic user traffic models, satellite scheduling scenarios, and automated network workflows, supporting fault-tolerant and delay/disruption-tolerant networking (DTN) studies. - Contributed to system-level integration, & verification of satellite link performance across LEO mega-constellations benchmark against real Starlink network.	
UAV Research Intern	Jan 2023 – July 2023
<i>Computational Intelligence Lab, Indian Institute of Science (IISc)</i>	<i>Bengaluru, India</i>
- Built autonomous system software and simulation pipelines for UAV-based sensing, integrating real-time data processing and system-level testing. - Built GNC modules with SPI & CAN-based radars & depth camera drivers for PX4 & K++ flight controllers, enabling robust system integration and operational reliability. - Designed ROS/Gazebo-based simulations for end-to-end mission workflows, improving canopy coverage efficiency by 40%.	
Guidance-Navigation-Control Intern	May 2022 – Aug 2022
<i>General Aeronautics Pvt Ltd</i>	<i>Bengaluru, India</i>
- Led the development of scalable SITL & HITL framework enhancing trajectory planning for the company's Sense & Avoid project, saving \$1000+ on expensive flight controllers & high-resolution sensors. - Implemented algorithms for real-time decision-making and fault-tolerant navigation, built SPI & CAN-based radar and depth camera sensor drivers, saving 3+ months of the control division's efforts. - Implemented a robust metaheuristic GNC algorithm with APF collision avoidance in SITL, enhancing efficiency by 20% and saving 100+ hours of ground testing.	

PROJECTS

Fault Tolerant Lunar C&PNT Architecture (Masters Thesis)	Jan 2025 – Apr 2026
- Developed a semi-deterministic Delay Tolerant Network (DTN) emulator for sustainable distributed and fault-tolerant lunar communication & PNT service studies validating compliance with LunaNet standards. - Built automated communication and navigation workflows using BPv7 protocol and TCP convergence layer for interoperable space network services. - Conducted performance evaluations of GNSS-like navigation infrastructure including least square & EKF solutions for real-time mission scenarios and distributed system simulations.	
Distributed Fault-tolerant Asteroid Inspection <i>MATLAB, Simulink, SPICE toolkit</i>	
- Constructed an integrated Adhoc SMC controller with stochastic MPC controller to incorporate unmatched uncertainties in asteroid dynamics under near-frozen orbit conditions for distributed asteroid inspection problem. - Built a swarm reconfiguration strategy that generates optimal satellite-target pairs and compute desired reference trajectories for the adaptive controller using collocation-based BVP formulation.	

1. **Aryan S.** et. al., LEO Satellite Network Testbed: MiniNet-based Emulation of Network Congestion and Routing Dynamics. *Acta Astronautica*. (In-Progress)
2. B. Barbour, Kedrowitsch A., Downs J., and Mhadgut D., **Aryan S.**, Black J., Kenyon S. "SpaceNet: A Resource-Efficient Simulation Testbed for LEO Mega-Constellation Networks," in *IEEE Access*, vol. 14, pp. 28097-28118, 2026, doi: 10.1109/ACCESS.2026.3666138. (IEEE Journal)
3. **Aryan, S.**, Downs J., Kedrowitsch A., Smith J., Houck A. and, Kenyon S. SpaceNet Testbed – Small Satellite Network Emulation of Real-World Scenarios. *DigitalCommons@USU - Small Satellite Conference 2025*, Salt Lake City, Utah. (Poster presentation - 3rd best student poster award)
4. **Aryan, S.** Reference tracking of nonlinear airborne systems using stochastic MPC with disturbance observers and actuator chance constraint optimization. *CEAS Aeronaut J* (2025). <https://doi.org/10.1007/s13272-025-00847-w> (Springer Nature journal)
5. Downs, J. & Barbour, B. & Kedrowitsch, A. & Mhadgut, D. & **Aryan, S.** & Kenyon, S. Space Network (SpaceNet) Testbed - Development of a Multi-Functional Testbed for Simulating Space Communication Networks. *AIAA Scitech 2025*. 10.2514/6.2025-2716. (Conference proceedings)
6. **Aryan, S.** & Fitzgerald, R. Four Body Invariant Structures And Chaos Analysis For Jovian Multi-Moon Ballistic Transfers. *AAS Astrodynamics Specialist conference 2024, Broomfield, CO*. (Conference proceedings)
7. More, D., Acharya, S., and **Aryan, S.**, "SRGAN-TQT, an Improved Motion Tracking Technique for UAVs with Super-Resolution Generative Adversarial Network (SRGAN) and Temporal Quad-Tree (TQT)," *SAE Technical Paper 2022-26-0021*, 2022, <https://doi.org/10.4271/2022-26-0021>. (Conference proceedings)

TECHNICAL SKILLS

Languages: Python, C/C++, C#, Julia, Markdown, HTML/CSS

Software Tools: MATLAB/Simulink, Mathematica, Ansys STK (certified), Robot Operating System (ROS), LabView, GMAT, KSP, FlightGear, QGC, MissionPlanner

DevOps & Systems Tools: Github/GitLab actions, Docker, DigitalOcean, VS Code, Visual Studio, PyCharm, Jupyter notebook, Google colab

Testing & Integration: Pytest, simulation (SITL /HITL)

Libraries: SciPy (signal processing), SPICE Toolkit, NumPy, Astropy, Poliastro, Pandas, Scikit, Matplotlib, SkyField, GEOS API, Tensorflow/Keras, OpenCV

Firmware/Hardware: Ardupilot, PX4, Betaflight, Navio2, Raspberry Pi Series, Jetson Nano, Intel NUC, CompactRIO